

Fraud Alert Control Tower

Explainable fraud alert prioritization, investigator queue control, threshold economics, and governance evidence.

Prepared by	Date	Status
Mirza Saif Baig	June 21, 2026	Submission-ready prototype using synthetic data

Executive summary

This project was built to solve an explainable fraud alert prioritization problem. I interpreted the brief as an operational analytics challenge: a fraud team cannot review every transaction, so the system must rank the riskiest cases, explain why each case is risky, and show how the threshold affects investigator workload.

Final conclusion

The project is not only a fraud classifier. It is a decision-support control tower that connects fraud scoring to review queues, investigator explanations, false positives, and governance controls.

Project journey

Phase	What I did	Why it mattered
Understand the brief	Reframed the task as fraud operations and review-capacity prioritization.	Made the work realistic for a banking fraud team.
Prepare the data	Joined transaction, customer, and merchant information; fixed known data issues; prevented leakage.	A model is only useful if the pipeline is trustworthy.
Build features	Selected the top 10 explainable features using training-only mutual information.	Kept the model defensible and app-friendly.
Compare models	Tested weighted models using fraud-focused metrics.	Accuracy is misleading when fraud is rare.
Deploy workflow	Built a Vercel-style control tower with scoring, queue, metrics, governance, and export.	Turned analysis into an operating workflow.

Data cleaning and preparation

The modelling table was created at transaction grain, with one row per transaction. Customer and merchant fields were joined in as supporting context. The cleaning stage focused on preserving records, flagging data quality issues, and removing fields that could leak the answer.

- Parsed transaction timestamps into hour, day-of-week, month, and weekend features.
- Normalized country values, including mapping GB to UK for consistency.
- Left joined customer and merchant profiles so transaction records were preserved.
- Created missing-profile flags rather than dropping unmatched customer records.
- Flagged broken tenure values instead of treating them as valid numeric tenure.
- Excluded IDs, target fields, and the existing alert flag from model inputs.

Feature engineering and selection

Feature selection used training-only mutual information. This avoided choosing features based on validation or test data and kept the final app aligned with the same inputs used by the modelling workflow.

Feature group	Examples	Reason
Location	geo_distance_km, txn_country	Captures unusual transaction location patterns.
Identity and device	synthetic_identity_score, device_risk_score	Adds account and device risk context.
Merchant	merchant_risk_score, merchant_profile_risk_score	Adds merchant-level fraud risk.
Transaction behavior	channel, txn_hour, amount, log amount	Captures channel, timing, and value behavior.

Modelling and threshold strategy

The modelling stage compared several weighted candidates because fraud was rare. I focused on ranking quality, precision, recall, false positives, and review count rather than accuracy alone.

Metric	Selected result	Interpretation
Recommended threshold	0.798	Selected around a 5% review-capacity assumption.
Review count	74 of 1,000 test transactions	Actual test queue rate was 7.4%.
Queue hit rate	13.5%	Share of reviewed cases confirmed as fraud.
Fraud capture	25.0%	Share of all frauds captured in the queue.
True positives	10	Fraud cases caught at this threshold.
False positives	64	Legitimate cases requiring human validation.

Threshold conclusion

The threshold is a business-control decision. Lower thresholds catch more fraud but increase false positives. Higher thresholds reduce workload but risk missing fraud.

Canonical risk-tier policy

Tier	Probability band	Analyst meaning
Critical	$p \geq \max(0.820, \text{review threshold})$	Emergency review before any customer-impacting decision.
High	$\text{review threshold} \leq p < \text{Critical band}$	Manual investigator review.
Medium	$0.350 \leq p < \text{review threshold}$	Monitor closely and escalate if behavior repeats or another signal appears.
Low	$p < 0.350$	Likely legitimate; keep score and reasons in the audit trail.

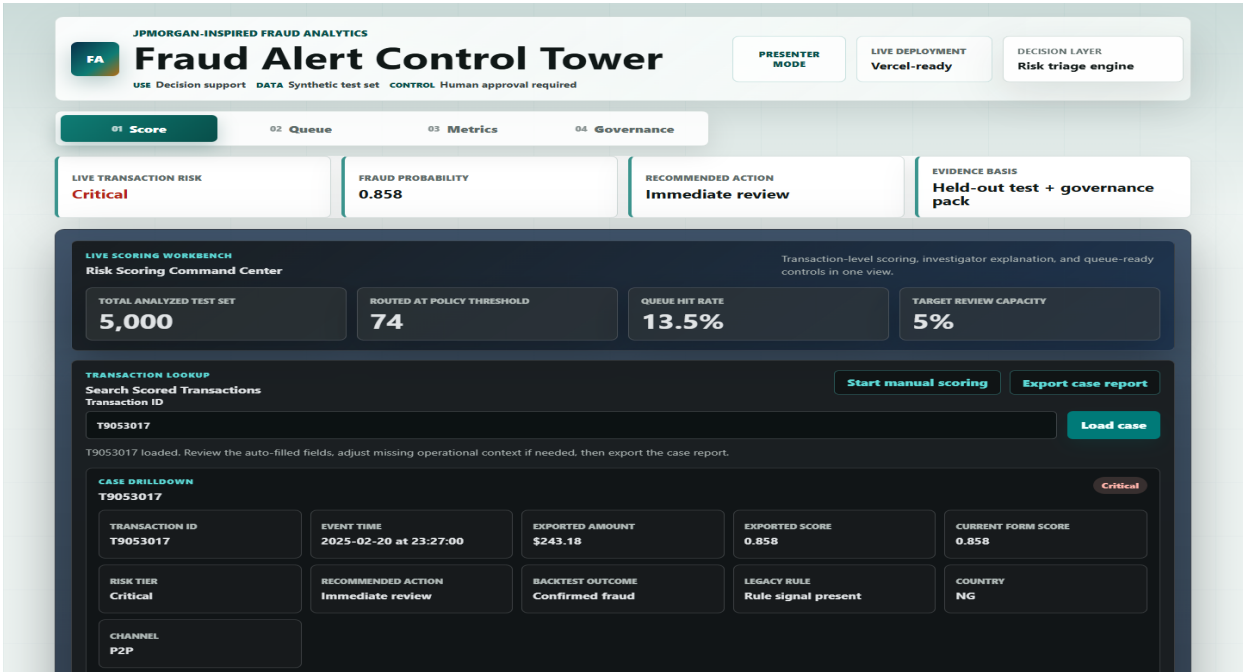
Business impact sensitivity

Item	Existing alert	Risk triage policy	Change
Reviewed cases	22	74	+52
Fraud cases caught	8	10	+2

Item	Existing alert	Risk triage policy	Change
False positives	14	64	+50
Illustrative net impact	-	\$584	Based on \$8 review cost and \$500 avoided loss

Deployed analyst app

The final app is a Vercel-style static application, so reviewers can open it in a browser without running a Python server. It contains Score, Queue, Metrics, and Governance views.



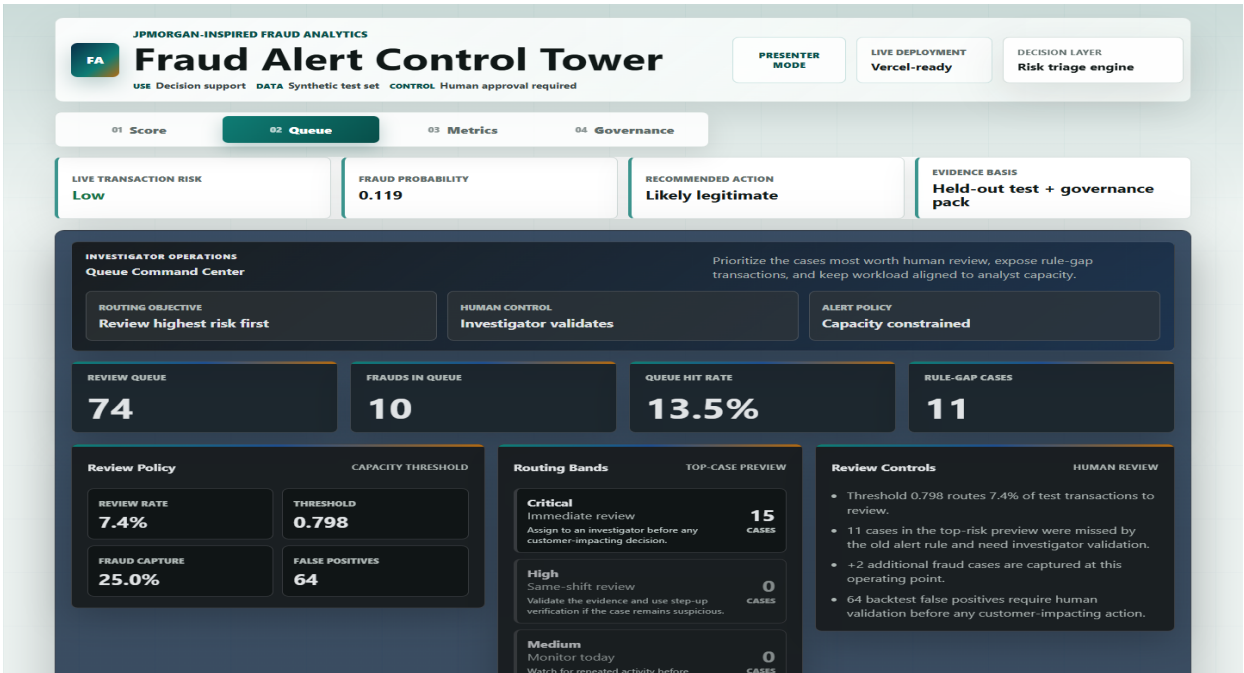
Score view: transaction lookup, manual scoring, risk decision, explanation, and CSV case export.

Section	Purpose
Score	Search or manually enter a transaction and receive risk probability, alert tier, action, explanation, and CSV export.
Queue	Show investigator review queue, top cases, rule-gap cases, triage lanes, and control checks.
Metrics	Translate model performance into fraud capture, false positives, review cost, and threshold trade-offs.
Governance	Summarize model card, limitations, approval gates, monitoring controls, and responsible-use policy.

Explainability and investigator use

The app explains the score by comparing each transaction with typical reference values. If a transaction has unusually high geographic distance or amount, the app shows how that signal increased model risk. If a feature lowers risk, the app presents it as a protective signal.

Important wording control: the explanation describes model behavior, not guaranteed causality. It means the model treated a feature value as risk-increasing or risk-reducing; it does not prove that the feature caused fraud.



Queue view: review load, frauds in queue, rule-gap cases, triage lanes, and control checks.

Governance and responsible use

Control	How it was handled
Leakage prevention	Removed IDs, target fields, and the old alert flag from model inputs.
Human review	App recommends review actions but does not automate blocking.
Threshold governance	Threshold selected with investigator capacity and false-positive load in mind.
Monitoring	Defined queue size, hit rate, fraud capture, false positives, drift, and segment review.
Limitations	Synthetic data means production use would require real-data validation and approval.

Changes made after tutor feedback

After receiving feedback, I improved the project to make it easier for a tutor or JPMorgan-style reviewer to understand and evaluate. The main changes focused on clarity, professional presentation, and reviewer usability.

Feedback theme	Change made
PPT needed a stronger story	Created a journey-based deck explaining how the project started, evolved, and reached its conclusion.
App needed clearer analyst flow	Improved opening state, labels, sections, and risk analyst wording.
Export needed to be more useful	Changed the case report export to CSV with transaction facts, decision evidence, and governance notes.
Tutor may ask how it was built	Added optional Presenter Mode so build notes appear only when needed.
Submission folder needed focus	Kept final submission materials project-focused and separated from learning notes.

Project report card

Review area	Evidence in this project
Fraud relevance and feature engineering	Top-10 fraud features, joins, leakage prevention, and explainable inputs.
Model evaluation and threshold discipline	Weighted model comparison, held-out metrics, capacity ladder, false-positive analysis.
Explainability and governance	Local explanations, model card, threshold memo, responsible-use summary, monitoring controls.
Operational dashboard usefulness	Score, Queue, Metrics, Governance, transaction lookup, drilldown, and CSV export.
Business communication	Alert economics, review cost framing, fraud capture trade-off, final PPT, and professional report.

Two-minute closing pitch

I built an explainable fraud alert prioritization system that cleans and joins transaction, customer, and merchant data; selects top risk features; compares fraud models; tunes the threshold around investigator capacity; and deploys a Vercel-style control tower with scoring, queue, metrics, governance, and case export.